

# AyurSamhita: A Collaborative Ayurvedic Telehealth & Integrative Medicine Platform

*An Integrative Medicine Tele-Consultation Ecosystem*

## 1. Project Overview & Executive Summary

**AyurSamhita** is an integrative, cloud-based telehealth platform designed to bridge the gap between traditional Ayurvedic medicine and modern healthcare practices. The application provides patients with personalized Ayurvedic wellness profiles (including Dosha analysis) and access to professional consultations, while simultaneously empowering healthcare professionals with specialized dashboards, digital prescription tracking, and a unique peer-to-peer **clinical collaboration module**.

By merging holistic diagnostics with real-time video teleconsultation (WebRTC) and multi-doctor collaborative consultation channels, AyurSamhita offers a unified treatment ecosystem for chronic disease management and holistic healing.

## 2. The Problem Statement

Despite the rising global demand for holistic and preventive healthcare, the industry faces three critical challenges:

- Lack of Integration:** Traditional Ayurvedic practitioners and modern Allopathic doctors operate in silos, preventing comprehensive, integrative care for patients with complex medical backgrounds.
- Impersonal Virtual Diagnostics:** Telehealth solutions rarely capture the holistic profile of a patient (such as lifestyle, dietary patterns, and constitutional factors like \*Dosha\*).
- Absence of Peer-to-Peer Collaboration:** Doctors lack secure, real-time channels to consult with specialist peers on patient cases, leaving cross-practice case discussions fragmented across insecure messaging channels.

## 3. The Solution: AyurSamhita

AyurSamhita addresses these gaps by establishing a secure, scalable web architecture that connects patients, primary care doctors, and secondary specialists:

- Holistic Diagnostics:** Patients identify their mind-body constitution via an automated Dosha assessment engine.
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**Integrative Knowledge Base:** Features a disease-to-treatment indexing directory comparing Ayurvedic herbs with Allopathic medicines.

- **Collaborative Medical Framework:** Allows primary doctors to directly invite other specialists into an appointment to collaborate on patient histories, current herb/medicine routines, and treatment plans.
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## 4. System Architecture & Technical Stack

### Frontend Architecture

- **Framework:** React.js with TypeScript (Vite bundler) for a type-safe, performant client.
- **UI/UX Styling:** Custom CSS/Vanilla Tailwind CSS implementing glassmorphism, responsive CSS Grid layouts, custom Micro-Animations, and modern typography (Google Fonts Outfit/Syne).
- **State Management & Routing:** React Context API (Auth Provider) & React Router Dom for role-based route protection.
- **Integrations:** Jitsi Meet iframe API & MiroTalk WebRTC interfaces for encrypted, browser-native video consultations.

### Backend Architecture

- **Runtime:** Node.js with Express.js REST API framework.
  - **Database:** MongoDB (NoSQL) using Mongoose ODM for schemas, flexible validation, and data modeling.
  - **Security & Auth:** Stateful session management, HTTP-only cookie-based authentication, and password hashing (bcrypt).
  - **Key Endpoints:** Role-Based Access Control (RBAC) separating `/patient-details`, `/doctors-bookings`, `/collaborate`, and `/search-doctors` routes.
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## 5. Core Modules & Features

### A. Patient Portal

- **Constitutional Analysis (Dosha Quiz):** Interactive diagnosis parser calculating Vata, Pitta, and Kapha distributions, generating personalized dietary and lifestyle recommendations.
- **Herb & Allopathic Browser:** Searchable, cross-indexed catalog helping users explore herbal remedies, side-effects, and modern equivalents for specific diseases.
- **Unified Appointment Scheduling:** Step-by-step booking system covering consultation type (Video vs. In-person), date-time slot reservation (linked to doctor availability databases), and symptom input.
- **Simulated Billing & Checkout:** Multi-channel mock checkout supporting Card, UPI, Wallets, and NetBanking with progressive loading animations.
- **Patient Consultation Hub:** Interactive meeting dialog to fetch secure, database-synced video links for active consultations.

## B. Doctor Portal

- **Clinical Dashboard:** Real-time analytical stats tracking total appointments, upcoming consults, completed cases, and peer-collaboration requests.
- **Patient EMR (Electronic Medical Records) Viewer:** Expandable profile viewer displaying patient blood groups, date of birth, medical histories, current medications, and saved herbal catalogs.
- **Peer-to-Peer Collaboration Console:** Action triggers permitting doctors to invite other specialists to consult on a patient's active booking file.

## C. Collaboration Engine

- Primary doctors can trigger peer search requests within the portal.
- Adding a collaborating doctor populates the patient's case into the collaborator's dashboard under a designated **"Collaborating Cases"** tab.
- The system allows multiple specialists to share clinical data, examine patient histories, and coordinate integrative treatments.

## 6. Key Technical Highlights & Engineering Decisions

1. **Date-Only Scheduling Alignment:** To solve time-zone drift issues commonly found in UTC date-objects, the scheduling engine strips the time coordinates for date-status matching. This ensures today's appointments display correctly as "Upcoming" rather than "Past" on both doctor and patient clients.
2. **Unified Peer Access Control (RBAC):** Implemented strict validation checks in the collaboration API, ensuring collaborating doctors can *\*only\** pull the patient's health details when actively invited to the case file, protecting user privacy.
3. **Database Population (Mongoose Refs):** Optimized complex queries using nested Mongoose `.populate()` paths to retrieve detailed user information through the `Doctor` models in a single database trip.

## 7. Future Scope & Scalability

1. **AI-Powered Diagnostics:** Integrating Machine Learning models to analyze tongue and facial features to complement the digital Dosha diagnostic quiz.
2. **Decentralized EMRs:** Utilizing blockchain frameworks to share patient data securely across Ayurvedic and Allopathic systems with cryptographic user consent.
3. **Real-time Prescriptions:** Adding an interactive prescription pad generator allowing doctors to export signed PDF prescriptions directly to the patient's dashboard.